MLOps Lab Assignment - 1 — B22BB038

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Links

- **WANDB visualizations:** View Assignment on WandB
- **GitHub Repository:** View Assignment on GitHub

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1. Objective

The objective of this lab was to:

- Train a Convolutional Neural Network (CNN) on the **CIFAR-10 dataset**.
- Implement a **custom PyTorch dataloader**.
- Compute **FLOPs** for the selected model.
- Train the model for **25 epochs**.
- Visualize **Gradient flow** and **Weight update flow**.
- Log all metrics and visualizations using **Weights & Biases (WandB)**.

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2. Dataset

We used the **CIFAR-10 dataset**, which contains:

- 60,000 RGB images of size **32×32**.
- 10 classes (airplane, car, bird, cat, etc.).
- 50,000 training images and 10,000 test images.

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3. Model Architecture

A **ResNet-18** model was chosen for its ability to handle deep feature extraction through residual blocks.

- **Key components:** Residual blocks, Batch normalization, and ReLU activations.

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4. Custom DataLoader Implementation

A custom `Dataset` class was implemented to manually handle data loading.

- **Augmentation:** Random crop, horizontal flip, and CIFAR-10 normalization.

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5. FLOPs and Complexity

Complexity was measured using the `thop` library:

- **Total FLOPs:** 37.22 Million (0.037 GFLOPs).
- **Total Parameters:** 11.18 Million.

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6. Training Setup

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	128
Epochs	25
Loss	CrossEntropyLoss

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7. Visualization Observations

Gradient Flow

- Gradients were monitored periodically via WandB bar charts.
- **Observation:** ResNet skip-connections maintained stable gradients across all 18 layers, preventing vanishing gradient issues.

Weight Update Flow

- Tracks the magnitude of change in parameters ($||W_{new} - W_{old}||$).
- **Observation:** Large updates occurred initially, with magnitude decreasing and stabilizing as the model converged towards 25 epochs.

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8. Results and Findings

- **Accuracy:** Training accuracy showed steady growth, reaching over 85% by the final epoch.
- **Validation:** Validation accuracy was tracked side-by-side to monitor generalization.
- **Convergence:** Model stability was achieved after ~15-20 epochs.